Ware2Door Database Design

1NF · 2NF · 3NF · BCNF · ER Diagram

1. Initial Schema

The Ware2Door schema was designed to support the complete warehouse-to-last-mile logistics workflow. Before normalisation, the schema consisted of the following eleven entities:

Entity	Purpose
Warehouse	Stores warehouse identification details, managerial contact information, and location data including geometric_coordinates.
Warehouse_Inventory	Tracks shipments stored and dispatched from warehouses.
TransitHub	Maintains hub details and managerial contact information for intermediate transit facilities.
Transit_Inventory	Records in-scan and out-scan operations for shipments passing through transit hubs.
LocalAgency	Stores agency information along with delivery agent details — agent_id, agent_name, agent_phone, and agent_email — all within the same table.
Orders	Captures order information received from third-party platforms, including customer details and the assigned warehouse.
Shipment	Links orders with warehouses, transit hubs, and local agencies; stores OTP codes, estimated delivery times, remarks, and delivery attempt counts.
LocalAgency_Inventory	Tracks shipments received at and dispatched from local agencies through in-scan and out-scan operations.
Assigned_Shipments	Tracks which shipments are assigned to which delivery agents and records delivery status updates.
Tickets	Stores customer support and operational issues linked to specific shipments.
RTO	Records return-to-origin requests raised when deliveries fail.

While the schema covered all required operational data, the review process identified three categories of issues that needed to be addressed through normalisation:

- **Composite attributes:** geometric_coordinates in Warehouse and TransitHub stored latitude and longitude as a single combined value rather than two independent fields.

- **Grouped attributes causing redundancy:** LocalAgency contained delivery agent details (agent_id, agent_name, agent_phone, agent_email) directly within the agency table, which would lead to data duplication if an agent was linked to multiple records.
- **Potential partial dependencies:** The presence of agent attributes inside LocalAgency under a composite key scenario created a structural risk of partial dependencies.

2. First Normal Form (1NF)

The first step was to ensure every attribute in the schema held a single, atomic value. Two composite attributes were identified and decomposed.

2.1 geometric_coordinates in Warehouse and TransitHub

Both Warehouse and TransitHub stored the physical location of the facility using a single geometric_coordinates attribute that bundled latitude and longitude into one value. Since these are two distinct and independently meaningful data points, they were separated into individual columns.

Before	After
geometric_coordinates	latitude
(e.g. "17.3850, 78.4867")	longitude

This change was applied to both Warehouse and TransitHub. With latitude and longitude stored separately, location-based queries and geospatial calculations can be performed directly without any string parsing.

After 1NF

- geometric_coordinates decomposed into latitude and longitude in both Warehouse and TransitHub.
- All remaining attributes across the eleven entities were verified to be atomic — no other composite fields were found.

The Ware2Door schema satisfies 1NF.

3. Second Normal Form (2NF)

With 1NF confirmed, the schema was checked for partial dependencies. This applies to any entity where non-key attributes do not depend on the full primary key. The issue was found in LocalAgency.

3.1 Delivery Agent Attributes in LocalAgency

In the initial design, LocalAgency contained delivery agent attributes — agent_id, agent_name, agent_phone, and agent_email — directly alongside the agency details. The implied composite

key was (agency_id, agent_id), but the agent's personal details depended only on agent_id and had no functional relationship with agency_id. This is a partial dependency.

Attribute	Depends On	Issue
agent_name	agent_id only	Partial dependency — does not depend on agency_id
agent_phone	agent_id only	Partial dependency — does not depend on agency_id
agent_email	agent_id only	Partial dependency — does not depend on agency_id

Keeping these attributes inside LocalAgency would mean that if an agent were associated with multiple agency records, their name and contact details would be duplicated across each row.

3.2 Resolution — Local_Delivery_Agent Extracted

The delivery agent attributes were extracted into a new, separate entity: Local_Delivery_Agent. This table uses agent_id as its sole primary key, and carries agency_id as a foreign key to maintain the relationship with LocalAgency. LocalAgency now holds only agency-level information.

LocalAgency — Before	After Split
agency_id (PK)	LocalAgency
agency_name	agency_id (PK)
address, city, state, pincode	agency_name
agent_id	address, city, state, pincode
agent_name	
agent_phone	Local_Delivery_Agent
agent_email	agent_id (PK)
	agency_id (FK → LocalAgency)
	agent_name
	agent_phone
	agent_email

Each agent's details are now stored exactly once. The agency relationship is preserved through the foreign key.

After 2NF

- Delivery agent attributes extracted from LocalAgency into a new Local_Delivery_Agent entity.

- agent_id is the sole primary key of Local_Delivery_Agent; agency_id is a foreign key.
- No other entities with composite keys were found to have partial dependencies.

The Ware2Door schema satisfies 2NF.

4. Third Normal Form (3NF)

After resolving partial dependencies, each table was examined to check whether any non-key attribute was determining another non-key attribute — a transitive dependency. Every non-key attribute across all entities was traced back to the primary key directly.

Entity	Primary Key	Transitive Dependency Found?
Warehouse	warehouse_id	No
Warehouse_Inventory	inventory_id	No
TransitHub	hub_id	No
Transit_Inventory	transit_inventory_id	No
LocalAgency	agency_id	No
LocalAgency_Inventory	la_inventory_id	No
Local_Delivery_Agent	agent_id	No
Orders	order_id	No
Shipment	shipment_id	No
Assigned_Shipments	assignment_id	No
Tickets	ticket_id	No
RTO	rto_id	No

After 3NF

- No transitive dependencies were found in any table.
- No changes to the schema were required at this stage.

The Ware2Door schema satisfies 3NF.

5. Boyce–Codd Normal Form (BCNF)

The final check verified that for every functional dependency in every table, the determinant is a superkey — meaning no attribute determines another unless it can uniquely identify a row on its own. All tables were reviewed against this condition.

Entity	Functional Dependency	Determinant is Superkey?
Warehouse	warehouse_id → all attributes	Yes
Warehouse_Inventory	inventory_id → all attributes	Yes
TransitHub	hub_id → all attributes	Yes
Transit_Inventory	transit_inventory_id → all attributes	Yes
LocalAgency	agency_id → all attributes	Yes
LocalAgency_Inventory	la_inventory_id → all attributes	Yes
Local_Delivery_Agent	agent_id → all attributes	Yes
Orders	order_id → all attributes	Yes
Shipment	shipment_id → all attributes	Yes
Assigned_Shipments	assignment_id → all attributes	Yes
Tickets	ticket_id → all attributes	Yes
RTO	rto_id → all attributes	Yes

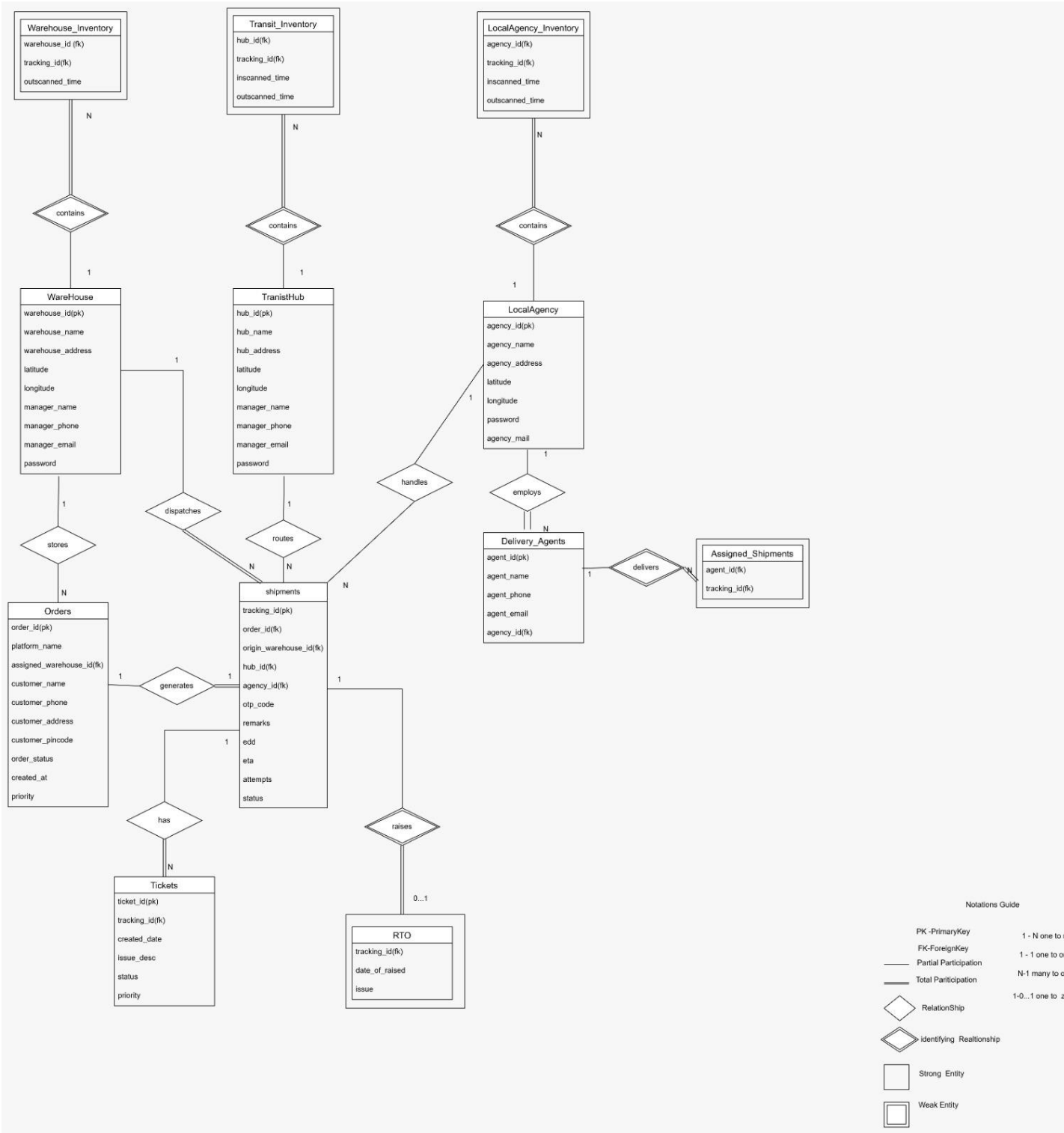
In every table, all functional dependencies are driven solely by the primary key, which is a superkey by definition. No non-key attribute determines any other attribute. There were no BCNF violations to resolve.

After BCNF

- All functional dependencies across all twelve entities are determined by superkeys.
- No further decomposition was required.

The Ware2Door schema is fully compliant with BCNF.

6. Entity-Relationship Diagram


Warehouse_Inventory
warehouse_id (fk)
tracking_id(fk)
outscanned_time
Transit_Inventory
hub_id(fk)
tracking_id(fk)
inscanned_time
outscanned_time
LocalAgency_Inventory
agency_id(fk)
tracking_id(fk)
inscanned_time
outscanned_time
contains
contains
contains
WareHouse
warehouse_id(pk)
warehouse_name
warehouse_address
latitude
longitude
manager_name
manager_phone
manager_email
password
TranistHub
hub_id(pk)
hub_name
hub_address
latitude
longitude
manager_name
manager_phone
manager_email
password
LocalAgency
agency_id(pk)
agency_name
agency_address
latitude
longitude
password
agency_mail
handles
employs
dispatches
routes
stores
Delivery_Agents
agent_id(pk)
agent_name
agent_phone
agent_email
agency_id(fk)
delivers
Assigned_Shipments
agent_id(fk)
tracking_id(fk)
Orders
order_id(pk)
platform_name
assigned_warehouse_id(fk)
customer_name
customer_phone
customer_address
customer_pincode
order_status
created_at
priority
shipments
tracking_id(pk)
order_id(fk)
origin_warehouse_id(fk)
hub_id(fk)
agency_id(fk)
otp_code
remarks
edd
eta
attempts
status
generates
has
raises
Tickets
ticket_id(pk)
tracking_id(fk)
created_date
issue_desc
status
priority
RTO
tracking_id(fk)
date_of_raised
issue
0...1
Notations Guide
PK -PrimaryKey
FK-ForeignKey
Partial Participation
Total Pariticipation
RelationShip
identifying Realtionship
Strong Entity
Weak Entity